

Problem Set 5: Cross-tabulation & Odds Ratio

Dataset A: Education Level vs Employment Status

Data (30 observations)

ID, EducationLevel, Employed

1, Graduate, Yes
2, Non-Graduate, No
3, Graduate, Yes
4, Graduate, No
5, Non-Graduate, No
6, Non-Graduate, Yes
7, Graduate, Yes
8, Non-Graduate, No
9, Graduate, Yes
10, Non-Graduate, Yes
11, Graduate, Yes
12, Non-Graduate, No
13, Graduate, No
14, Graduate, Yes
15, Non-Graduate, Yes
16, Non-Graduate, No
17, Graduate, Yes
18, Graduate, Yes
19, Non-Graduate, No
20, Non-Graduate, Yes
21, Graduate, No
22, Graduate, Yes
23, Non-Graduate, Yes
24, Non-Graduate, No
25, Graduate, Yes
26, Non-Graduate, Yes
27, Graduate, Yes
28, Graduate, No
29, Non-Graduate, No
30, Non-Graduate, Yes

Questions:

1. Construct a 2×2 contingency table of EducationLevel (Graduate / Non-Graduate) vs Employed (Yes / No).
2. Calculate the percentage employed within the Graduate group, and within the Non-Graduate group. Interpret the difference.

3. Compute the odds of being employed for Graduates and for Non-Graduates. Then compute the odds ratio (Graduate vs Non-Graduate). Interpret the result.
4. Under the hypothesis of independence, compute the expected cell counts and compare them with the observed counts. Does there appear to be an association?
5. What does an odds ratio greater than 1 imply in this context? What would an odds ratio of 1 imply? What if it were less than 1?
6. Discuss any caveats when interpreting these results (e.g., small sample size, possible confounders).

Dataset B: Residence Type vs Clean Water Access

Data (30 observations)

ID,ResidenceType,CleanWaterAccess

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1,Urban,Yes
2,Urban,Yes
3,Rural,No
4,Rural,Yes
5,Urban,Yes
6,Rural,No
7,Urban,No
8,Rural,No
9,Urban,Yes
10,Rural,Yes
11,Urban,Yes
12,Rural,No
13,Rural,Yes
14,Urban,Yes
15,Rural,No
16,Urban,No
17,Rural,Yes
18,Urban,Yes
19,Rural,No
20,Urban,Yes
21,Rural,Yes
22,Urban,No
23,Rural,Yes
24,Rural,No
25,Urban,Yes
26,Urban,Yes
27,Rural,No
28,Urban,Yes
29,Rural,Yes
30,Urban,No
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Questions:

1. Create the contingency table for ResidenceType (Urban / Rural) $\times$ Clean WaterAccess (Yes / No).
2. What percentage of urban households have clean water access? What about rural households? Interpret.
3. Compute the odds of having clean water access for urban vs rural households, and compute the odds ratio (Urban vs Rural). Interpret.
4. Under independence, compute expected cell-counts and compare with observed counts. What does this tell you?
5. What does an odds ratio > 1 , $= 1$ or < 1 imply in this context?
6. Suggest one possible confounder (e.g., income level, local governance) and discuss how it might affect interpretation.

Dataset C: Gender vs Participation in Political Meetings

Data (30 observations)

ID,Gender,AttendedPoliticalMeeting

1,Male,Yes
2,Female,No
3,Male,Yes
4,Female,Yes
5,Male,No
6,Female,No
7,Male,Yes
8,Female,No
9,Male,Yes
10,Female,Yes
11,Male,No
12,Female,No
13,Male,Yes
14,Female,Yes
15,Male,No
16,Female,No
17,Male,Yes
18,Female,Yes
19,Male,No
20,Female,No
21,Male,Yes
22,Female,Yes
23,Male,Yes

24,Female,No
25,Male,Yes
26,Female,No
27,Male,No
28,Female,Yes
29,Male,Yes
30,Female,No

Questions:

1. Form the contingency table for Gender (Male / Female) $\times$ AttendedPoliticalMeeting (Yes / No).
2. Calculate the percentage of males who attended a political meeting, and of females. Interpret.
3. Compute the odds of attendance for males vs females and compute the odds ratio (Male vs Female). Interpret the result.
4. Compute expected counts under independence, compare to observed. What does this suggest?
5. Interpret what an odds ratio > 1 , $= 1$, or < 1 means in this context.
6. Discuss limitations of this sample/survey (e.g., who was surveyed, what counts as a “political meeting”, gender norms) and how they might affect interpretation.